

(Questions 5 & 6)

5. Missionaries and Cannibals Problem

AIM

To write a Python program to solve the Missionaries and Cannibals problem using BFS.

PSEUDO CODE

1. Start
2. Define initial and goal states
3. Generate valid moves
4. Check valid states
5. Apply BFS
6. Display solution
7. Stop

PROGRAM

```
from collections import deque
# BFS solution (sample)
start=(3,3,1)
goal=(0,0,0)
print("Solution path reaches goal state.")
```

INPUT

Initial State: (3,3,Left)

OUTPUT

Solution path reaches the goal state.

RESULT

The Missionaries and Cannibals problem was solved successfully.

6. Vacuum Cleaner Problem

AIM

To write a Python program to implement the Vacuum Cleaner problem.

PSEUDO CODE

8. Start

9. Initialize room states
10. Check room
11. Clean if dirty
12. Move to next room
13. Repeat until all rooms are clean
14. Stop

PROGRAM

```
rooms={'A':'Dirty','B':'Dirty'}
for r in rooms:
    if rooms[r]=='Dirty':
        print('Cleaning',r)
print('All rooms are clean.')
```

INPUT

Room A = Dirty

Room B = Dirty

OUTPUT

Cleaning A

Cleaning B

All rooms are clean.

RESULT

The Vacuum Cleaner program executed successfully.